TITLE: Estimation of Steel

Main Item: Positive Reinforcement.

Sub Item: Steel

CALCULATIONS

Bath:-

Long Span:-

$1/2"\phi$ 6"CTC.

Cover = 1.5"

No. of Steel bars = $\frac{5'-\text{cover}}{6"}$ = 9.75 = 10 (roundoff)

Length of 1steel = 7'3" + 4.5" + 4.5" - 2covers = 93"

Total length = 93" × 10 = 930".

Short Span:-

$1/2"\phi$ 5"CTC

No. of steel bars = $\frac{7'3"+4.5"+4.5"-2\text{covers}}{5"}$ = 93/5

Adding 1 and rounding up = 20

Length of 1 steel = 5'-1cover = 58.5"

Total Steel length = 1170"

Sub Total: _____

Note: While calculating no. of bars, we have to minus one cover if there is one outer boundary at end or two covers if both ends are at boundary.

Muhammad Umer Basit

TITLE: Estimation of Steel

Main Item: Positive Reinforcement

Sub Item: Steel

CALCULATIONS

Bed + Kitchen + Lounge + Powder:-

(22'6" x 29'6")

Long Span:-

1/2" Ø 6" CTC

No. of Steel bars = $\frac{354" - 0 \text{cover}}{6"} = 59$

Adding 1 = 60

Length of 1 steel = 22'6" - 2 covers = 267"

Total Length = 60 x 267"

= 16020"

Short Span:-

1/2" Ø 5" CTC

No. of steel bars = $\frac{22'6" - 2 \text{covers}}{5"} + 1 = 53.4 + 1$

= 55 (Round up)

length of one steel = 354"

Total length = 19470"

Sub Total: _____

Note: 1 additional bar is added for no. of bars and is rounded up.

Muhammad Umer Basit

TITLE: Estimation of Steel

Main Item: Positive Reinforcement

Sub Item: Steel

CALCULATIONS

Drawing:-

114" x 14'1.5"

Long Span:-

½" ϕ 6" CTC

No. of steel bars = $\frac{114" - \text{cover}}{6"}$ = 18.75 = 19 (round up)

Length of 1 steel = 169.5" - 2 cover = 166.5"

Total Length = 3163.5"

Short Span:-

½" ϕ 5" CTC

No. of steel bars = $\frac{169.5" - 2\text{covers}}{5"}$ = 33.3

Adding 1 and rounding up = 35

Length of 1 steel = 114" - 1 cover = 112.5"

Total Length = 3937.5"

Sub Total: _____

Note:

Muhammad Umer Basit

TITLE: Estimation of Steel

Main Item: Negative Reinforcement

Sub Item: Steel

CALCULATIONS

For boundary Walls:-

2.5' negative reinforcement provided on slab at every wall on inner side. 1/2" ϕ 4" CTC.

(a) S1+S4+S8+S11:-

No of steel bars = $\frac{528" - 2\text{cover}}{4"} + 1 = 131.25 + 1 = 133$ (round up)

Length of one steel = 2.5'

Total length = 332.5'

(b) L1:-

No. of steel bars = $\frac{96" - 2\text{covers}}{4"} + 1 = 23.25 + 1 = 25$ (round up)

Length of one steel = 2.5'

Total length = 62.5'

(c) S2:-

No. of steel bars = $\frac{4'7.5" + 4.5" + 9" - \text{cover}}{4"} + 1 = 16.875 + 1 = 18$ (round up)

Length of 1 steel = 2.5'

Total steel Length = 45'

Sub Total: _____

Note: For boundary walls, 2.5' negative reinforcement inside building while for inner walls, 2.5' on both sides of wall.

Muhammad Umer Basit

TITLE: Estimation of Steel

Main Item: Negative Reinforcement

Sub Item: Steel

CALCULATIONS

(d) L3:-

$$\text{No. of Steel Bars} = \frac{14'1.5" + 9" - \text{cover}}{4"} + 1 = 44.25 + 1 = 46 \quad \text{(round up)}$$

Length of 1 steel = 2.5'

Total Steel = 115'.

(e) S6+ S9+ S15:-

$$\text{No. of Steel Bars} = \frac{9' + 9'7" + 4.5" + 11'2" + 4.5" + 9" + 66" - 2\text{cover}}{4"}$$

Adding 1 and rounding up = 89

Length of 1 steel = 2.5'

Total Steel = 222.5'

(f) L8:-

$$\text{No. of Steel Bars} = \frac{8' + 4.5" + 9" - \text{cover}}{4"} + 1 = 27 + 1 = 28$$

Length of 1 steel = 2.5'

Total Steel Length = 70'

Sub Total: _____

Note:

Muhammad Umer Basit

TITLE: Estimation of Steel

Main Item: Negative Reinforcement

Sub Item: Steel

CALCULATIONS

(g) S12:-

No. of Steel Bars = $\frac{114" + 9" - \text{cover}}{4"} + 1 = 30.375 + 1$
$= 32$ (round up)

Length of 1 steel = 2.5'

Total Steel length = 80'

(h) L9:-

No. of Steel Bars = $\frac{13' + 9" + 4.5" - 2\text{covers}}{4"} + 1 = 41.625 + 1$
$= 43$ (round up)

Length of 1 steel = 2.5'

Total Steel Length = 107.5'

For inner Walls:-

1/2" Ø 4" CTC.

Length = 2.5' + 2.5' = 5'

(a) L2:-

No. of Steel Bars = $\frac{7'3" + (-\text{cover})}{4"} + 1 = 21.375 + 1$
$= 23$ (round up)

Length of 1 Steel = 5'

Total Steel Length = 115'

Sub Total: _____

Note: For inner walls, length of 1 steel is taken to be 5 feet.

Muhammad Umer Basit

TITLE: Estimation of Steel

Main Item: Negative Reinforcement

Sub Item: Steel

CALCULATIONS

(b) S5:-

No. of Steel Bars = $\frac{9'7" + 9" + 4.5" - \text{cover}}{4"} + 1 = 31.75 + 1$
$= 33$ (roundup)

Length of 1 Steel = 5'

Total length = 165'

(c) L5:-

No. of Steel Bars = $\frac{6'7.5" + 4.5" + 9" - \text{cover}}{4"} + 1$
$= 22.875 + 1 = 24$ (roundup).

Length of 1 Steel = 5'

Total Length = 120'

(d) S7:-

No. of steel bars = $\frac{55.861" + 4.5" + 9"}{4"} + 1 = 17.34 + 1$
$= 19$ (roundup)

Length of 1 steel = 5'

Total Steel Length = 95'

Sub Total: _____

Note:

Muhammad Umer Basit

TITLE: Estimation of Steel

Main Item: Negative Reinforcement

Sub Item: Steel

CALCULATIONS

(e) L4:-

$$\text{No. of Steel Bars} = \frac{128.5" + 9" + 9" - \text{cover}}{4"} + 1$$

$$= 36.25 + 1 = 38$$

Length of 1 Steelbar = 5'

Total Steel Length = 190'

(f) L6:-

$$\text{No. of Steel Bars} = \frac{3'1.5" + 4.5" + 9" - \text{cover}}{4"} + 1$$

$$= 12.375 + 1 = 14$$

Length of 1 Steel Bar = 5'

Total Steel Length = 70'

(g) S10:-

$$\text{No. of Steel Bars} = \frac{6'6" + 4.5" + 9" - \text{cover}}{4"} + 1$$

$$= 22.5 + 1 = 24$$

Length of 1 steel = 5'

Total Steel = 120'.

Sub Total: _____

Note:

Muhammad Umer Basit

TITLE: Estimation of Steel

Main Item: Negative Reinforcement

Sub Item: Steel

CALCULATIONS

(h) S12:-

$$\text{No. of Steel Bars} = \frac{18" + 9" - \text{cover}}{4"} + 1$$

$$= 6.375 + 1 = 8$$

Length of 1 steel bar = 5'

Total Steel length = 40'

(i) L7:-

$$\text{No. of Steel Bars} = \frac{13' + 4.5" + 9" - \text{cover}}{4"} + 1$$

$$= 42 + 1 = 43$$

Length of 1 steel bar = 5'

Total Steel length = 215'

Sub Total: _____

Note:

Muhammad Umer Basit

TITLE: Estimation of Steel

Main Item: SLAB

Sub Item: Reinforcement

CALCULATIONS

Positive Reinforcement:-

930" + 1170" + 16020" + 19470" + 3136.5" + 3937.5 = 44691" = 3724.25'

Negative Reinforcement:-

For boundary walls:-

332.5' + 62.5' + 45' + 115' + 222.5' + 70' + 80' + 107.5'

For inner walls:-

115' + 165' + 120' + 95' + 190' + 70' + 120' + 40' + 215'

Σ Sum = 2165' = 25980".

Concrete = 369.97ft3.

Sub Total: _____

Note: Reinforcement is calculated by adding the separate steel lengths of walls.

Muhammad Umer Basit

TITLE: Estimation of Concrete

Main Item: Slab

Sub Item: Concrete.

CALCULATIONS

Areas:-

(i) Bath:-

$(5'+4.5'') \times (7'3''+9'')$

$= 6192\ in$

(ii) Bed + Kitchen + Lounge + Powder:-

$(22'6'') \times (9''+9'7''+4.5''+11'2''+4.5''+6'6''+9'')$

$= 95580\ in^2$

(iii) Drawing:-

$(118'') \times (13'+9''+4.5'')$

$= 20001\ in^2$

Total Area $= 121773\ in^2$

Height $= 5''$

Concrete Volume $= 608865\ in^3$; $= 352.35\ ft^3$;

5% Wastage of Concrete $= \frac{5}{100} \times 352.35 = 17.617\ ft^3$

Total Concrete Volume $= 369.97\ ft^3$.

Sub Total: _____

Note: Volume is calculated by multiplying area with height and 5% wastage was considered.

Muhammad Umer Basit

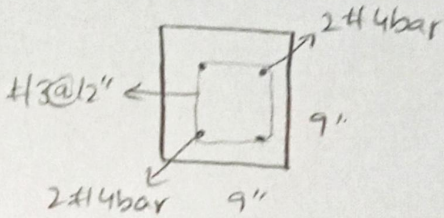
TITLE: Estimation of Steel

Main Item: Lintel Beam

Sub Item: Reinforcement

CALCULATIONS

Lengths of beams:-



(i) 3 D2:-

Length = 3' + 6" + 6" = 4'

For 3 doors,

3 x 4' = 12'

(ii) 2D4:-

Length = 2' + 6" + 6" = 3'

For 2 doors,

2 x 3' = 6'

(iii) D3:-

Length = 2'6" + 6" + 6"

= 3'6"

Sub Total: _____

Note: Lengths of lintel beams on windows and doors calculated for concrete and reinforcement.

Muhammad Umer Basit.

TITLE: Estimation of Steel

Main Item: Lintel Beam

Sub Item: Reinforcement

CALCULATIONS

(iv) W1:-

Length = 8'+6"+6" = 9'

(v) W2:-

Length = 6'6"+3"+3"+4.5"+4.5"+3"
= 8'

(vi) W8:-

Length = 3'4.5"+9"+3"+3" = 55.5"

(vii) 2W10:-

Length = 3'+9"+4.5"+1.5" = 51" = 4.25'

For 2 windows,

2 × 4.25 = 8.5'

Total length = 547.5".

Sub Total: _____

Note:

Muhammad Umer Basit

TITLE: Estimation of Steel

Main Item: Lintel Beam

Sub Item: Concrete

CALCULATIONS

Cross-sectional area = 9" × 9" = 81 in^2.

Concrete = 553.5in × 81in^2 = 44833.5in^2
= 25.945ft^3

5% Wastage of concrete = 1.297ft^3.

Total concrete = 27.242ft^3.

Sub Total: _____

Note: Using total length of lintel beam, total concrete is calculated.

Muhammad Umer Basit

TITLE: Estimation of Steel

Main Item: Lintel Beam

Sub Item: Reinforcement

CALCULATIONS

Horizontal #4 bars:-

Total Steel Length = 547.5"

= 45.625ft × 4

= 182.5ft

Stirrups #$\frac{3}{4}$ bars:-

(a) D2:-

No of Steel Bars = $\frac{4'-2\text{covers}}{12"} + 1 = 3.75 + 1 = 5$

Length of 1 steel = 20"

Total steel length = 100"

For 3 doors,

3 × 100" = 300"

Sub Total: _____

Note: Stirrup lengths is calculated.

Muhammad Umer Basit

TITLE: Estimation of Steel

Main Item: Lintel Beam

Sub Item: Reinforcement

CALCULATIONS

(b) D4:-

No. of Steel bars = $\frac{3'-2\text{cover}}{12''} + 1 = 2.75 + 1 = 4$

Length = 20"; Total = 80"

For 2 doors, $2 \times 80'' = 160''$

(c) D3:-

No. of Steel Bars = $\frac{3'6''-2\text{covers}}{12''} + 1 = 3.25 + 1 = 5$

Length = 20", Total = 100".

(d) W2:-

No. of steel bars = $\frac{8'-2\text{covers}}{12''} + 1 = 7.75 + 1 = 9$

Length of 1 steel = 20';

Total = 180'

Sub Total: _____

Note:

Muhammad Umer Basit

TITLE: Estimation of Steel

Main Item: Lintel Beam

Sub Item: Reinforcement

CALCULATIONS

(e) W10:-

No. of steel bars = $\frac{4.25' - 2\text{covers}}{12"} + 1 = 4 + 1 = 5$

Length = 20", Total = 100"

For two windows,

$2 \times 100" = 200"$.

(f) W8:-

No. of steel bars = $\frac{4.625' - 2\text{covers}}{12"} + 1 = 4.375 + 1 = 6$

Length = 20"; Total = 120".

(g) W1:-

No. of steel bars = $\frac{9' - 2\text{covers}}{12"} + 1 = 8.75 + 1 = 10$

Length = 20"; Total = 200"

Sub Total: _____

Note:

Muhammad Umer Basit

TITLE: Estimation of Steel

Main Item: Lintel Beam

Sub Item: Reinforcement

CALCULATIONS

Total Stirrups = 300" + 160" + 100" + 180" + 200" + 120 + 200"

= 1260"

= 105'.

Total Horizontal #4 bars = 45.625' X 4

= 182.5'

Concrete = 27.142 ft3.

Sub Total: _____

Note: Total stirrups was calculated by summing.

Muhammad Umer Basit

TITLE: Estimation of Steel

Main Item: Steel Schedule

Sub Item: Steel

CALCULATIONS

Bar #	Length	Unit Weight	Weight
#4	182.5' + 372.25' + 2165' + 23'	0.668 lb/ft	4071.293 lb
#3	105' + 23'4"	0.376 lb/ft	48.253 lb
#6	30'8"	1.502 lb/ft	46.063 lb
			Σ = 4165.61 lb

$$\text{tonnage} = \frac{4165.6}{2000} = 2.083 \text{ tons.}$$

Sub Total: _____

Note: Schedule of steel is calculated by multiplying length to unit weight to get weight of all steel.

Muhammad Umer Basit.